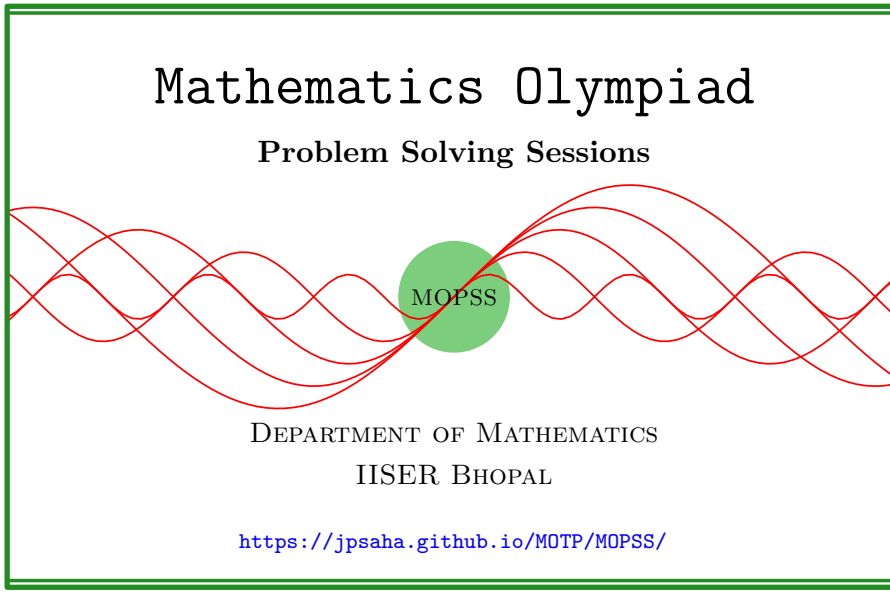
MOPSS

23 August 2025



Suggested readings

- Evan Chen's advice On reading solutions, available at https://blog.evanchen.cc/2017/03/06/on-reading-solutions/.
- Evan Chen's Advice for writing proofs/Remarks on English, available at https://web.evanchen.cc/handouts/english/english.pdf.
- Notes on proofs by Evan Chen from OTIS Excerpts [**Che25**, Chapter 1].
- Tips for writing up solutions by Edward Barbeau, available at https://www.math.utoronto.ca/barbeau/writingup.pdf.
- Evan Chen discusses why math olympiads are a valuable experience for high schoolers in the post on Lessons from math olympiads, available at https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/.

Exercise 1 (IOQM 2024 P8,). Let n be the smallest integer such that the sum of the digits of n is divisible by 5, as well as the sum of the digits of $n+1$ is divisible by 5. What are the first two digits of n in the same order?

Summary — Consider the last digit of n. Next consider its second last digit, the third last digit etc.

Walkthrough —

(a) Show that the last digit of n is 9.

(b) Note that n has at least two digits. If n is a two-digit number, then show that it is not equal to any integer other than 19 or 69.

(c) Conclude that n has at least three digits, and its second last digit is 9.

(d) If n is a three-digit number, then show that it is not equal to any integer other than 299 or 799.

(e) Argue similarly to show that n has at least four digits, and its third last digit is 9.

(f) Prove that n has at least five digits, and its fourth last digit is 9.

(g) If n is a five-digit number, then show that it is equal to 49999.

Do we need to check that 49999 has the required property?

Solution 1. ■

Exercise 2 (IOQM 2024 P10,). Determine the number of positive integral values of p for which there exists a triangle with sides a, b and c which satisfy

$$a^2 + (p^2+9)b^2 + 9c^2 - 6ab - 6pbc = 0$$

Summary — Complete squares and use the triangle inequality.

Walkthrough —

(a) Note that

$$a^2 + (p^2+9)b^2 + 9c^2 - 6ab - 6pbc = (a-3b)^2 + (pb-3c)^2.$$

(b) This gives $a = 3b$ and

$$c = \frac{p}{3}b.$$

(c) Use the inequality[a] $a + b > c$, to obtain

$$3b + b > \frac{p}{3}b,$$

which yields $p \leq 11$.

The content posted here and at this blog by Evan Chen are quite useful.

(d) Use the inequality[b] $b + c > a$, to get

$$b + \frac{p}{3}b > 3b,$$

implying $p \geq 7$.

(e) It reduces to finding the number of positive integers p such that $7 \leq p \leq 11$.

What about the condition $c + a > b$? Anything else to be taken care of?

[a]How does it follow?
[b]How does it follow?

Solution 2. ■

Exercise 3 (IOQM 2024 P15,). Let X be the set consisting of twenty positive integers $n, n+2, \ldots, n+38$. The smallest value of n for which any three numbers $a, b, c \in X$, not necessarily distinct, form the sides of an acute-angled triangle is:

Summary — Use Pythagoras' theorem.

Walkthrough —

(a) Use the fact that[a] if a, b, c are the sides of an acute-angled triangle with c denoting the largest side, then $a^2 + b^2 > c^2$ holds.

(b) Show that for any such n, the inequality

$$(n + 38)^2 < 2n^2$$

holds, which yields $n \geq 92$.

(c) Prove that for $n = 92$, the inequality

$$2(n + 2i)^2 > (n + 2i + 2)^2$$

holds for $i = 0, 1, \ldots, 18$.

(d) Note that if suffices to show that

$$2m^2 > (m + 2)^2$$

for any $m \in \{92, 94, \ldots, 128\}$.

What is the role of part (c)? Does it guarantee that for $n = 92$, the set X has the stated property?

[a]How to prove it? Does the **converse** of this statement hold? In other words, if a, b, c are positive real numbers satisfying $a^2 + b^2 > c^2$, then does it follow that a, b, c are

the sides of an acute-angled triangle? Even if it doesn't, would it be true under additional hypothesis?

Solution 3. ■

Exercise 4 (INMO 2025 P5, , proposed by Pranjal Srivastava and Rohan Goyal)**.** Greedy goblin Griphook has a regular 2000-gon, whose every vertex has a single coin. In a move, he chooses a vertex, removes one coin each from the two adjacent vertices, and adds one coin to the chosen vertex, keeping the remaining coin for himself. He can only make such a move if both adjacent vertices have at least one coin. Griphook stops only when he cannot make any more moves. What is the maximum and minimum number of coins that he could have collected?

Walkthrough —

(a) Show that 1998 coins can be collected (see Fig. 1).

(b) Show that the difference of the number of coins at the vertices with odd indices, and the number of coins at the vertices with even indices, does not change modulo 3 after any move.

(c) Conclude that no more than 1998 coins can be collected.

(d) Show that 668 coins can be collected.

(e) Show that at least 668 coins are collected when Griphook stops.

The following is from AoPS, and is due to Rohan Goyal.

Solution 4. Denote the regular 2000-gon by $A_1A_2\dots A_{2000}$. The claims below imply that the maximum (resp. minimum) number of coins that Griphook could have collected is 1998 (resp. 668).

Claim — No more than 1998 coins can be collected.

Proof of the Claim. Note that after any move, the number of coins at the vertices with odd indices and the number of coins at the vertices with even indices either reduces by 2 and increases by 1 respectively, or increases by 1 and reduces by 2 respectively. This shows that the difference of the number of coins at the vertices with odd indices and the number of coins at the vertices with even indices, does not change modulo 3 after any move, and hence, this difference is zero modulo 3 after any move.

Since the number of coins remaining after any move is at least 1, there are at least two coins left at the end of any move. Consequently, no more that 1998 coins can be collected. □

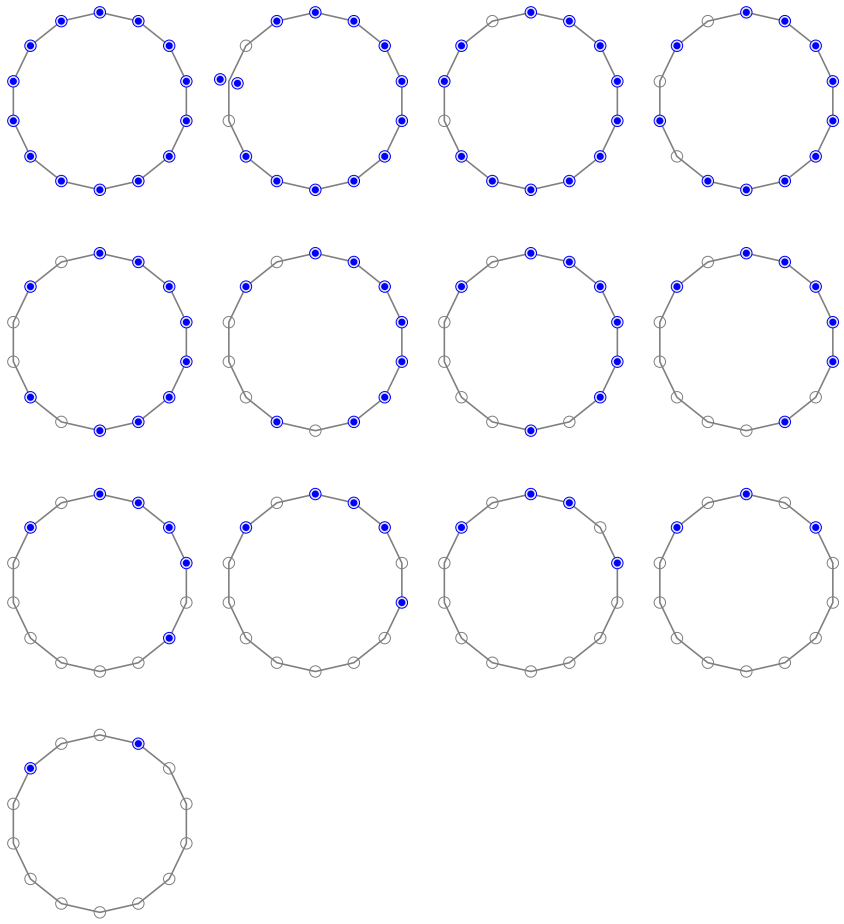


Figure 1: INMO 2025 P5, Collecting 12 coins, Exercise 4

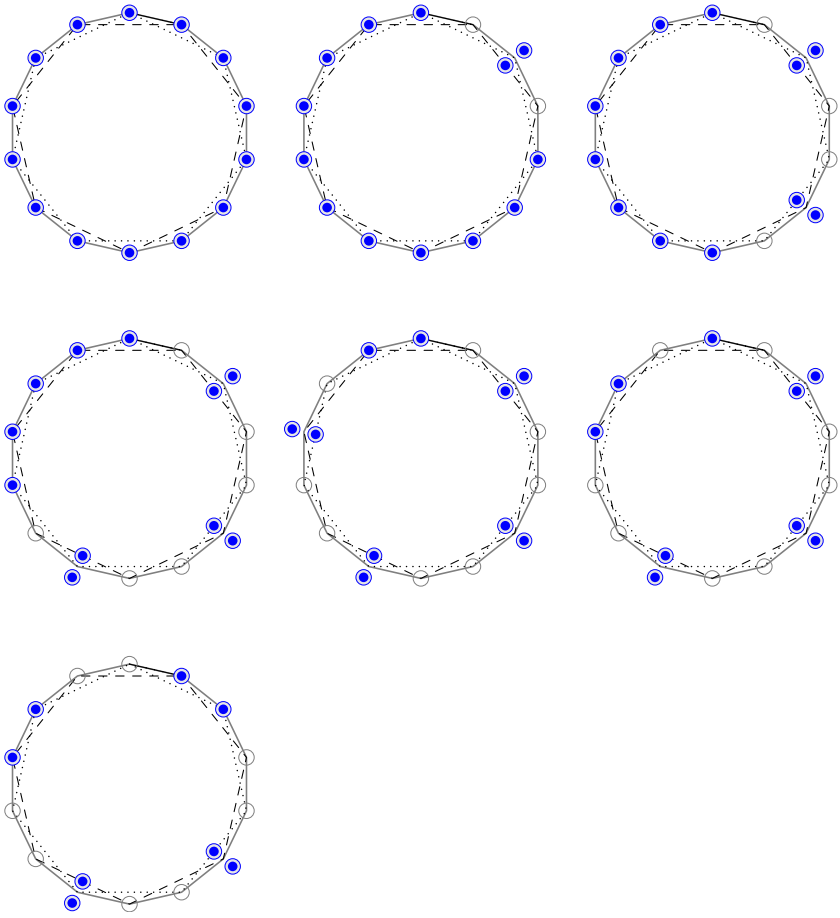


Figure 2: INMO 2025 P5, Collecting 6 coins, Exercise 4

Claim — There is a way to collect 1998 coins.

Proof of the Claim. Let us denote the number of coins at the vertices of the regular 2000-gon by a string of length 2000, consisting of nonnegative integers, where the i-th entry of the string denotes the number of coins at the i-th vertex A_i.

In the following, 1^m (resp. 0^m) denotes a string of m 1's (resp. 0's). Note that by making a move, the string 1^{2000} can be turned to $1^{1996}0201$, which can be turned to $1^{1995}01010$ by making another move. Observe that the sub-string 101 can be turned to 010 by making a move. Hence, making enough suitable moves, the string $1^{1995}01010$ can be reduced to $010^{1996}10$. While reaching $010^{1996}10$ from 1^{2000}, a total of 1998 coins have been earned, proving the claim. □

Claim — There is a sequence of moves, which leads to the collection of 668 coins when Griphook stops.

Proof of the Claim. Write the string 1^{2000} as $1(111)^{666}1$, which can be turned to $1(020)^{666}1$, which is equal to $1020(020)^{664}0201$. This can be reduced to $0110(020)^{664}0110$. Note that no further move is possible, and hence Griphook has to stop, and he earned $2000 - 4 - 2 \times 664 = 668$ coins. □

Claim — For any sequence of moves, Griphook will have at least 668 coins after he stops.

Proof of the Claim. Let c_i denote the number of coins at the i-th vertex. Note that each move reduces the sum

$$S = \sum_{i=1}^{2000} \min\{c_{i-1}, c_{i+1}\}.$$

In this sum, the indices are to be read modulo 2000. In each move, this sum reduces at least by 1, and Griphook stops precisely when this sum reduces to 0. Note that at any moment, for any two one-vertex apart vertices, one of them contains at most one coin[1], that is,

$$\min\{c_{i-1}, c_{i+1}\} \leq 1$$

for any $1 \leq i \leq 2000$. Write

$$E = \sum_{1 \leq i \leq 2000, 2 \mid i} \min\{c_{i-1}, c_{i+1}\},$$

[1]Why?

$$O = \sum_{1 \le i \le 2000, 2 \nmid i} \min\{c_{i-1}, c_{i+1}\}.$$

Note that in each move, one of E, O is reduced at most by 3. Hence, at least 334 moves are required to reduce any of E, O to 0. Since in each move, only one of E, O gets reduced, it follows that Griphook makes at least $334 + 334 = 668$ moves before he stops.

□

■

Exercise 5 (USAJMO 2013 P1, proposed by Titu Andreescu)**.** Are there integers a, b such that $a^5b + 3$ and $ab^5 + 3$ are both perfect cubes of integers?

Summary — Consider the integer ab modulo 3, and read $a^5b + 3, ab^5 + 3$ modulo 9.

Walkthrough —

(a) If 3 divides ab, then consider one of the integers $a^5b + 3, ab^5 + 3$ modulo 9.

(b) If 3 does not divide ab, then assuming the integers $a^5b + 3, ab^5 + 3$ to be perfect cubes, determine the integers a^5b, ab^5 modulo 9.

(c) Consider the integer $(ab)^6$ modulo 9.

Solution 5. Let us assume that there are integers a, b satisfying the given condition. First, let us consider that 3 divides ab. Without loss of generality[2], let us assume that 3 divides a. Then

$$a^5b + 3 \equiv 3 \pmod 9.$$

Since $a^5b + 3$ is a perfect cube and is divisible by 3, it follows that

$$a^5b + 3 \equiv 0 \pmod 9,$$

implying $3 \equiv 0 \pmod 9$, which is impossible.

Next, let us consider the case that 3 does not divide ab. Since $a^5b + 3$ is a cube and is not divisible by 3, we obtain $a^5b + 3 \equiv \pm 1 \pmod 9$, which implies that a^5b is congruent to one of $5, 7$ modulo 9. Using a similar argument, it follows that ab^5 is also congruent to one of $5, 7$ modulo 9. Hence, $(ab)^6$ is congruent to one of $5 \cdot 5, 5 \cdot 7, 7 \cdot 7$ modulo 9. So, $(ab)^6$ is not congruent to 1 modulo 9. Since 3 does not divide ab, it follows that $(ab)^6$ is congruent to 1 modulo 9.

It follows that there are no integers a, b satisfying the given conditions. ■

[2]Show that there is indeed no loss of generality in assuming what follows.

Exercise 6 (USAMO 2000 P4,). Find the smallest positive integer n such that if n squares of a 1000×1000 chessboard are colored, then there will exist three colored squares whose centers form a right triangle with sides parallel to the edges of the board.

Walkthrough —

(a) Does Fig. 3 help?

(b) Consider a suitable coloring to show that $n \geq 1999$.

(c) Does coloring 1999 squares suffice?

(d) What happens when there are two columns, each containing at least two colored squares, and some row contains one colored square from each of these columns?

(e) Consider the columns which contain at least two colored squares. How many colored squares can they contain together?

(f) What about the remaining colored squares? How many are they in total?

Solution 6. Note that if all the squares on the top row and on the left column, excluding the square common to them, are colored, then the required condition is not satisfied. This shows that n is at least 1999.

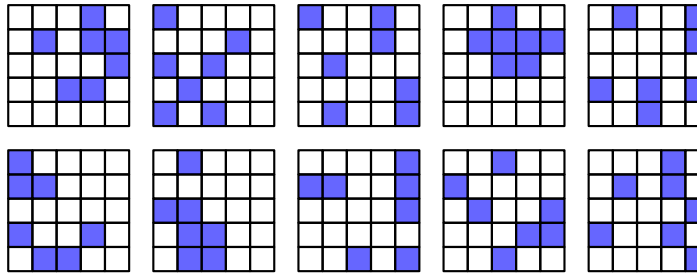
Suppose precisely 1999 squares of the chessboard have been colored. Note that there are columns containing at least two colored squares. Consider the columns containing at least two colored squares. If there are two among these columns, such that some row contains one colored square from each of these columns, then we are done. It remains to consider the case that no two of these columns have this property, which we assume from now on. Hence, the colored squares lying on these columns, lie in different rows. This shows that these columns together contain at most 1000 colored squares. Consequently, there are at least 999 colored squares, each lying on a colum containing no more colored squares. This implies that there are precisely 999 such colored squares. Moreover, there is only one column containing at least two colored squares, and it contains the remaining 1000 colored squares. One among the 999 colored squares, together with two suitable squares from these 1000 squares, have the required property. This shows that $n = 1999$. ■

Exercise 7 (USAMO 2018 P4, , proposed by Ankan Bhattacharya). Let p be a prime, and let $a_1, a_2, \ldots, a_p$ be integers. Show that there exists an integer k such that the numbers

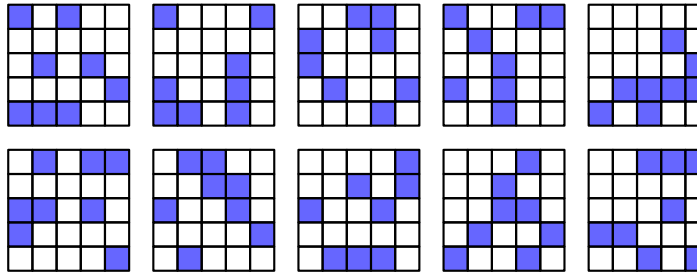
$$a_1 + k, a_2 + 2k, \ldots, a_p + pk$$

produce at least $p/2$ distinct remainders upon division by p.

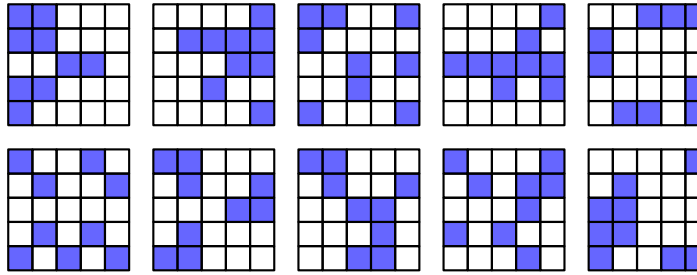
The following uses the fact that every integer not divisible by a prime p is invertible modulo p, which can be proved from first principles!



The above chessboards have 7 of its squares colored



The above chessboards have 8 of its squares colored



The above chessboards have 9 of its squares colored

Figure 3: USAMO 2000 P4, Exercise 6

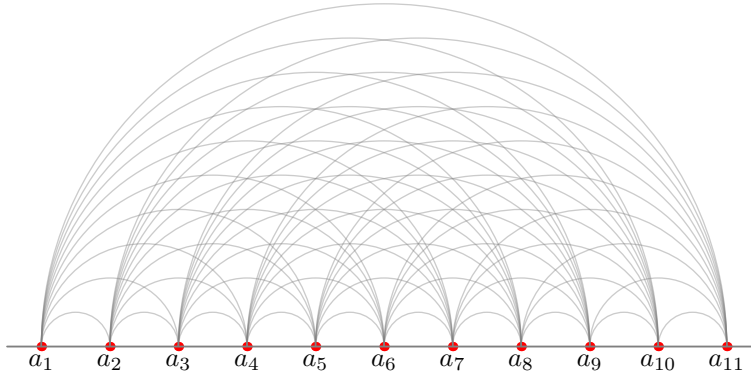


Figure 4: USAMO 2018 P4, Exercise 7

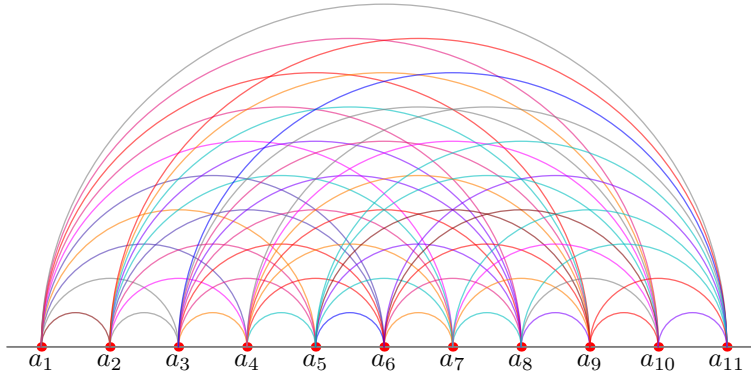


Figure 5: USAMO 2018 P4, Exercise 7

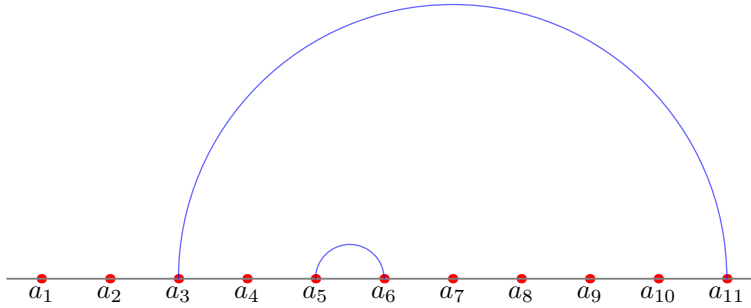


Figure 6: USAMO 2018 P4, Exercise 7

Walkthrough —

(a) Note that it suffices to show that some integer k among $1, 2, \ldots, p$ has the required property.

(b) Consider the integers $a_1, a_2, \ldots, a_p$, and we join each of them using an arc as in Fig. 4.

(c) For any two distinct integers i, j among $1, 2, \ldots, p$, show that the i-th and the j-th ones among the numbers

$$a_1 + k, a_2 + 2k, \ldots, a_p + pk,$$

that is, the numbers $a_i + ik, a_j + jk$ leave the same remainder upon division by p precisely for one value of k.

(d) Let us color the arcs (see Fig. 4). We join a_i, a_j by an arc if the numbers $a_i + ik, a_j + jk$ leave the same remainder upon division by p. We write k on that arc. See Fig. 5, where instead of writing the corresponding integers on the arcs, we have colored them. Thinking of the integers $1, 2, \ldots, p$ as p colors, and we have *colored* the arcs instead of *labelling* the arcs[a].

(e) Note that there are precisely $\frac{p(p-1)}{2}$ unordered pairs of the form $\{a_i, a_j\}$, that is, there are precisely $\frac{p(p-1)}{2}$ arcs (see Fig. 4). Moreover, on each of these arcs, one of the integers $1, 2, \ldots, p$ is written. In Fig. 5, these arcs are colored using at most p colors.

(f) Conclude that for some integer k among $1, 2, \ldots, p$, at most $(p-1)/2$ arcs have the integer k written on them, that is, there are at most $(p-1)/2$ unordered pairs of the form $\{a_i, a_j\}$ such that $a_i + ik, a_j + jk$ produce the same remainder upon division by p. In Fig. 5, for some color, there are at most $(p-1)/2$ arcs of that color.

(g) Show that[b] for this integer k, the numbers

$$a_1 + k, a_2 + 2k, \ldots, a_p + pk$$

produce at least $p/2$ distinct remainders upon division by p.

[a] Observe that all the p colors might not have been used, for instance, if any two of $a_1, \ldots, a_p$ differ by a multiple of p.

[b] A graph on n vertices with m edges has at least $n - m$ connected components. This can be proved by induction on m, and observing that introducing a new edge reduces the number of connected components at most by 1.

References

[**Che25**] Evan Chen. *The OTIS Excerpts*. Available at https://web.evanchen.cc/excerpts.html. 2025, pp. vi+289 (cited p. 1)